

NEUROLOGICAL EMERGENCIES

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○ Objectives

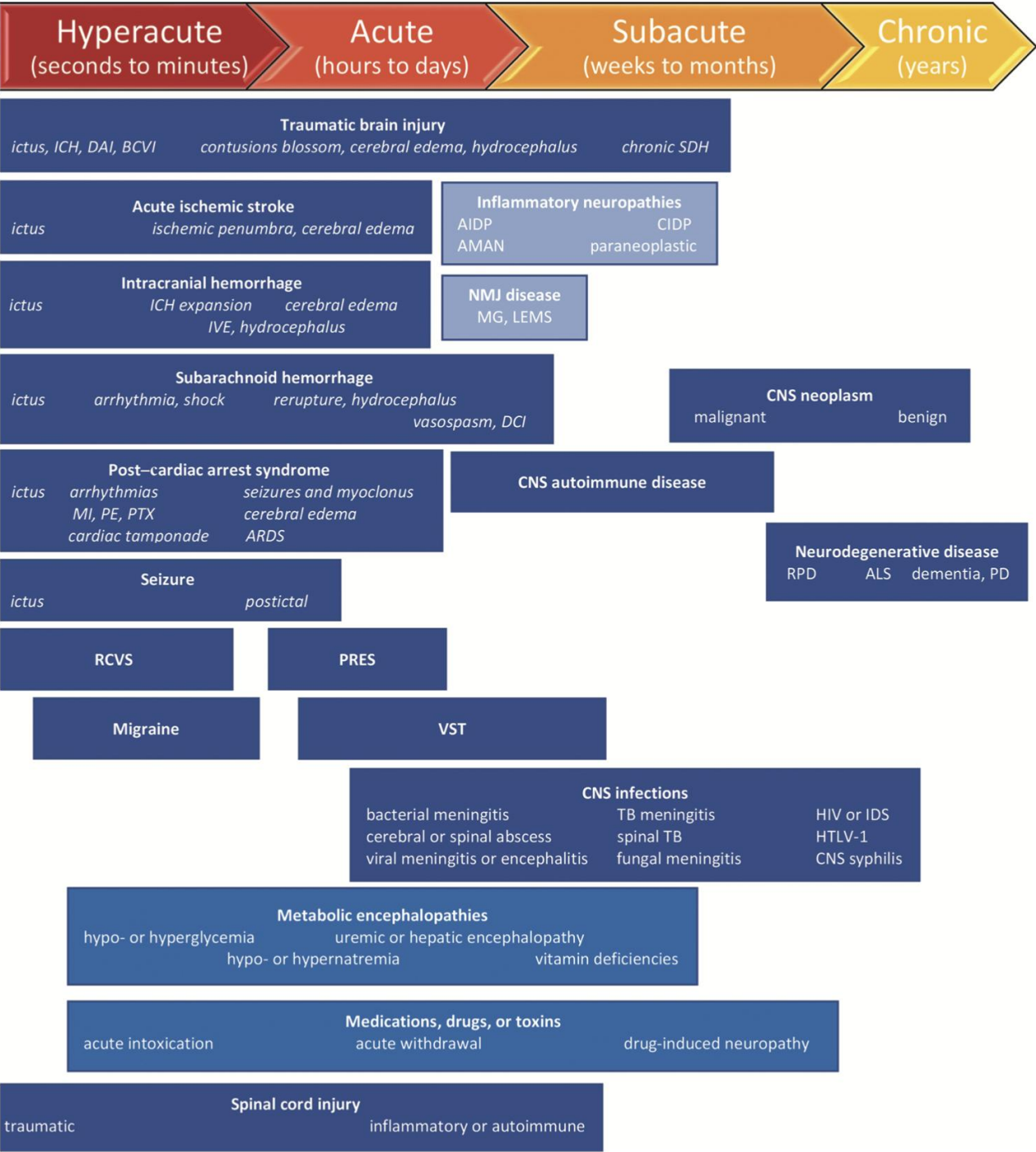
- Be able to generate a differential diagnosis for neurological emergencies based on patient presentation
- Know red flags in focused history for neurological complaints
- Understand how to perform Glasgow Coma Scale exam
- Be able to define and differentiate coma vs minimally conscious state vs persistent vegetative state
- Understand the 20/30/40 criteria for intubation
- Understand the significance of last known well time in ischemic strokes
- Be able to identify timing of interventions for acute ischemic stroke
- Be able to differentiate different types on hemorrhagic strokes on CT scan images
- Understand utility of lumbar puncture in diagnosis of SAH
- Be able to define status epilepticus and its initial management based on stage and severity
- Know gold standard tests for diagnosis of acute ischemic stroke, hemorrhagic stroke, status epilepticus, spinal cord injury, neuromuscular emergencies



○ Intro to neurological emergencies

- Approximately 5% of all ED patients present with neurological complaints
- Common complaints: confusion, headache, dizziness, weakness, back pain, and seizure
- **Acute neurologic emergencies** are common and often difficult to diagnose
- *Prompt recognition is crucial.* Delays can have devastating consequences with long-term disability
- Important to quickly recognize/initiate treatment even before neurologic consultation
 - Critical time windows for therapeutic interventions to improve neurologic function or prevent secondary neurologic injury
- Potential for rapid, life-threatening deterioration, often manifesting with changes in the neurologic examination
- **Time is Brain**





○ Approach to neurological emergencies

- Goal:

1. Identify the **imminent threats** to the patient: Vitals are vital, ABCs

1. *Respiratory failure*: airway monitoring for GCS <8

2. *Hemodynamic compromise*: blood pressure → cerebral perfusion

2. **Recognize** and **localize** the problem

1. Differential diagnosis built on **focused history and clinical examination**

2. What is the problem?: symptom time course, progression, and evolution

3. Where is the problem?: neurologic examination localization of underlying process on the neuraxis

3. Consider need for **emergent interventions**

1. Problem is relatively clear and time sensitive i.e. stroke

2. Problem is not clear and less time sensitive: pursue more comprehensive workup



○ History: getting the story

- Critical step in differential diagnosis and guiding initial workup
- Frequently gathered in parallel to workup in *hectic pace*
- Often need collateral info: family, EMS, other clinicians, and chart
- **LAST KNOWN WELL** – time of onset is crucial
 - Hyperacute (seconds to minutes), acute (hours to days), subacute (weeks to months), chronic (years)
 - Evolution of the symptoms: stable, fluctuating, progressive, episodic
- PMH/PSH, medications, etc all provide context
- High morbidity and mortality – goals of care



○ History taking: red flags

1. Rapid decline in **level of consciousness**
2. New confusion, behavioral changes, or agitation
3. New **motor** weakness or **sensory** deficit
4. New language or speech disturbance
5. New inability to **walk or balance** difficulties
6. New **visual** symptoms
7. New involuntary movements or **posturing**
8. New-onset **severe headaches**



○ Can't miss diagnoses

Localization	Underlying condition	Common acute threats and critical concerns
Brain ^a	Traumatic brain injury	Mass lesion or intracranial hemorrhage (ICH), diffuse cerebral edema leading to elevated intracranial pressure (ICP) and herniation; blunt cerebrovascular injury causing ischemic stroke; facial trauma, airway compromise; polytrauma (pneumothorax, perforation, penetrating injuries); long bone fractures leading to cerebral fat emboli syndrome; hydrocephalus; paroxysmal sympathetic hyperactivity, autonomic instability
	Acute ischemic stroke	Large vessel occlusion causing worsening cerebral ischemia; insufficiently perfused penumbra leading to worsening cerebral ischemia; malignant cerebral edema causing herniation (middle cerebral artery, cerebellum), hydrocephalus (cerebellum); cardiovascular comorbidities (heart failure, cardiac arrhythmias, intracardiac thrombus)
	Spontaneous intracranial hemorrhage	ICH expansion; mass effect leading to elevated ICP and herniation; intraventricular extension of ICH leading to hydrocephalus
	Subarachnoid hemorrhage	Aneurysmal rerupture; cardiac arrhythmias, hemodynamic instability; hydrocephalus; postprocedure (coil or clip) complications; neurogenic cardiomyopathy, cardiogenic shock; cerebral vasospasm; delayed cerebral ischemia
	Post-cardiac arrest syndrome and hypoxemic ischemic encephalopathy	Recurrent cardiac arrhythmias and cardiac arrest; underlying arrest causes (eg, myocardial infarctions, pulmonary embolism, pneumothorax, cardiac tamponade); hemodynamic instability; acute respiratory distress syndrome (ARDS); complications of therapeutic hypothermia; seizures and myoclonus; cerebral edema causing elevated ICP and herniation
	Venous sinus thrombosis	Venous congestion leading to worsening edema, ICH, and herniation; underlying systemic causes of hypercoagulable state
	Status epilepticus	Permanent neurologic injury; ictal cardiac arrhythmias; side effects of antiepileptic medications and sedatives
	Posterior reversible encephalopathy syndrome (PRES)	Worsening and permanent neurologic deficit; cerebral ischemia and ICH; complications related to underlying cause (eclampsia, fetal distress, transplant rejection)





Localization	Underlying condition	Common acute threats and critical concerns
Brain ^a	Central nervous system (CNS) infections	Mass effect or edema causing elevated ICP and herniation; vasculitis, ischemic and hemorrhagic strokes; hydrocephalus; complications related to systemic infection
	CNS malignancies	Mass effect or edema causing elevated ICP and herniation; cerebral ischemia and ICH; hydrocephalus; coagulopathy; complications related to systemic malignancy
	Autoimmune CNS disorders	Neurologic deterioration; vasculitis, ischemic and hemorrhagic strokes; side effects of immunosuppressive medications, infection
	Toxidromes	ECG changes, cardiac arrhythmias; seizures, withdrawal; serotonin syndrome, neuroleptic malignant syndrome, malignant hyperthermia
Spinal cord	Traumatic spinal cord injury	Cord compression (external [bone], internal [syrinx]); hemodynamic instability (cervicothoracic cord), bradycardia, asystole; hypoventilation, inadequate secretion management, diaphragmatic weakness leading to acute respiratory failure
	Spinal cord infections, malignancies, autoimmune or inflammatory disease	Hemodynamic instability (cervicothoracic cord); bradycardia, asystole
Peripheral nerve	Acute inflammatory demyelinating polyradiculoneuropathy (AIDP), acute motor axonal neuropathy (AMAN)	Hypercarbic respiratory failure, respiratory acidosis; autonomic instability, cardiac arrhythmias
Neuromuscular junction	Myasthenia gravis or botulism	Hypercarbic respiratory failure, acidosis, myasthenic crisis, cholinergic crisis

^a Common concerns in all acute brain injuries are acute respiratory failure due to insufficient airway protection and seizures in patients with cortical pathology.



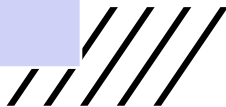
○ Examination: general principles

- Guides localization and differential ddx: Central or peripheral nervous system pathology?
 - CNS and PNS pathologies can coexist
- Often limited by patient's inability to cooperate due to impaired level of consciousness
 - Brainstem reflexes, eye position, and motor responses to noxious stimuli are critical
- Examination may rapidly change – frequent exams needed
- Multiple confounders must be considered
 - Medications, sedation, metabolic derangements, systemic illness, etc



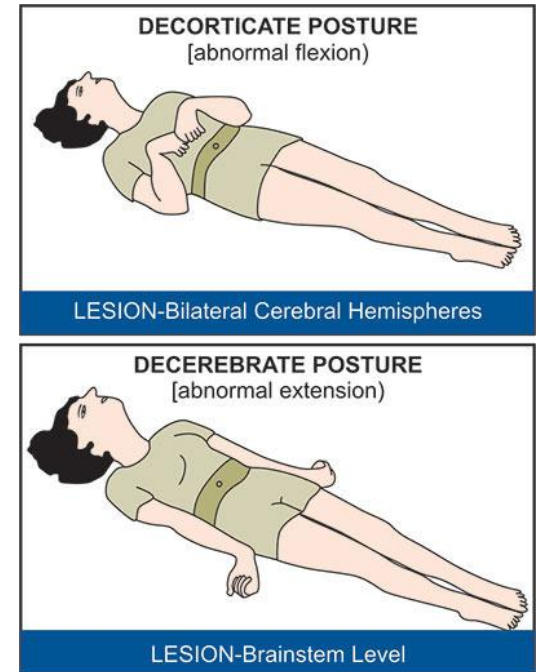
○ Definitions: level of consciousness

	Consciousness	Motor	Auditory	Visual	Communication
Coma	None	Reflexes only	None	None	None
Persistent Vegetative State	Arousal, no awareness	Posturing, withdrawal, not purposeful	Startle	Startle	None
Minimally Conscious State	Partial	Localizing, reaching, scratching	Localize sound	Fixation and pursuit intact	Inconsistent



○ CNS Exam: Acute Brain Injury

- Priority: look for signs of elevated ICP, stroke syndromes, seizures
 - These require **prompt intervention** – craniotomy, thrombectomy, thrombolysis, sedation etc. Time is Brain.
- Exam is tailored to level of consciousness
 - Test for response to vocal, tactile, and then noxious stimuli
- Gaze deviations
- Brainstem reflexes: blink, pupillary, corneal, oculoccephalic, oculovestibular, gag, cough
- Motor response – often first sign of acute deterioration
- Posturing – *decorticate vs decerebrate*
- Regular (seizure, tremor) and irregular movements (myoclonus, shivering)
 - Generalized or focal?



○ Herniation Syndromes



Type of herniation	Neuroanatomical pathology	Clinical signs
1 Uncal or lateral transtentorial herniation	Temporal lobe medializes toward and below the tentorium causing midbrain and third nerve compression; posterior cerebral artery (PCA) compression; may result in an indentation of the contralateral cerebral peduncle against the tentorium (Kernohan notch)	Ipsilateral fixed and dilated pupil, contralateral hemiparesis, coma; PCA syndrome; ipsilateral hemiparesis with Kernohan phenomenon
2 Central transtentorial herniation	Downward displacement of the diencephalon and brainstem causing compression of the hypothalamus and reticular activating system	Coma, bilateral small pupils; decorticate then decerebrate posturing; rostral then caudal loss of brainstem reflexes; Parinaud syndrome (loss of upgaze, retraction nystagmus); diabetes insipidus
3 Subfalcine or cingulate herniation	Displacement of the cingulate gyrus, corpus callosum (and ipsilateral anterior cerebral artery [ACA] or pericallosal artery) under the falx; compression of the contralateral foramen of Monro	Coma; ACA syndrome (contralateral leg weakness); dilation of the contralateral ventricle
4 Transcalvarial herniation	Cerebral displacement through fracture or surgical skull defect	Based on the location of the skull defect and underlying cortex
5 Cerebellar upward or ascending transtentorial herniation	Increased pressure in the posterior fossa, pushing the midbrain upward through the tentorial notch; PCA and superior cerebellar artery (SCA) compression	Cerebellar dysfunction, pupillary abnormalities; hydrocephalus (anterior brainstem displacement); PCA and SCA syndrome
6 Cerebellar downward or tonsillar herniation	Cerebellar tonsils move downward via the foramen magnum causing brainstem, cerebellar, fourth ventricular compression; posterior inferior cerebellar artery compression	Head tilt, neck stiffness, cerebellar dysfunction; severe headaches (if conscious) followed by coma; hemodynamic instability (often hypertension); cardiorespiratory dysfunction, cardiac arrest; Wallenberg syndrome

○ Central Nervous System Exam: Spinal Cord Injury

- Acute spinal Cord injuries: traumatic or rapidly progressive
- Start with **level** and type of injury based on motor and sensory function
- Reduced or absent rectal tone – early clue
- Loss of bowel/bladder function
- Autonomic dysfunction (neurogenic shock)
- Acute areflexia (early on) followed by increased tone and hyperreflexia (traditional upper motor neuron injury sign)
- **Lesions above C5:** diaphragmatic weakness – need to assess for intubation
- **Lesions above T6:** loss of sympathetic/vasomotor tone resulting in peripheral vasodilation – Shock!



○ Peripheral Nervous System Exam

- Acute Polyradiculoneuropathies (AIDP, Guillain Barre)
- Neuromuscular junction disorder (myasthenia, botulism)
- Critical task: **identify respiratory failure** and need for intubation
- Intubation Criteria: 20/30/40 rule
 - **Vital capacity <20ml/kg** (air exhaled after maximal inhalation)
 - Maximum inspiratory pressure aka **negative inspiratory force/NIF** worse than **-30 cm H2O** (inhale maximally, hold 1-2 sec)
 - **Maximum expiratory pressure <40cm H2O** (exhale maximally, hold 1-2 sec)



Scales and Scores:

Scale or score	Population	Components
Clinical examination		
American Spinal Cord Association Impairment Scale ⁴²	Spinal cord injury	Complete or incomplete motor and sensory function
Erasmus Guillain-Barré Syndrome Respiratory Insufficiency Score ³⁸	Guillain-Barré syndrome	Days between weakness and hospitalization, facial or bulbar weakness, Medical Research Council sum score
Full Outline of UnResponsivness (FOUR) score ⁴³	Mixed neurocritical care population	Motor response, eye response; brainstem reflexes, respiration
Glasgow Coma Scale (GCS) ⁴⁴	Described in traumatic brain injury, often applied to other subtypes of acute brain injury	Eye-opening: 4 = spontaneous, 3 = to sound, 2 = to pain, 1 = none Verbal response: 5 = orientated, 4 = confused, 3 = inappropriate, 2 = incomprehensible, 1 = none Motor response: 6 = obeys commands, 5 = localizes, 4 = flexion withdrawal, 3 = abnormal flexion, 2 = extension, 1 = none
Hughes Functional Grading Scale ⁴⁵	Guillain-Barré syndrome	Ability to run or walk, requiring a walker, requiring mechanical ventilation
Hunt and Hess Scale ⁴⁶	Aneurysmal subarachnoid hemorrhage (SAH)	Level of consciousness, headaches, nuchal rigidity, focal neurologic deficits
National Institutes of Health Stroke Scale ^{5,47}	Acute ischemic stroke	Level of consciousness; comprehension; speech; language; gaze; visual symptoms; face, arm, and leg weakness, sensory symptoms; ataxia; extinction
Quantitative myasthenia gravis score ^{48,49}	Myasthenia gravis	Upward (ptosis) and lateral gaze, facial muscles, swallowing, speech, vital capacity, upper extremity strength
World Federation of Neurological Surgeons grading scale ⁵⁰	Aneurysmal SAH	GCS, presence of motor deficit

Glasgow Coma Scale

EYE OPENING		VERBAL RESPONSE		MOTOR RESPONSE	
					
Spontaneous	4	Oriented	5	Obeys commands	6
To sound	3	Confused	4	Localizing	5
To pressure	2	Words	3	Normal flexion	4
None	1	Sounds	2	Abnormal flexion	3
		None	1	Extension	2

GLASGOW COMA SCALE SCORE

Mild 13–15	Moderate 9–12	Severe 3–8
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Objective measures, helpful over time and different examiners



○ CNS Imaging

- CT scan non contrast: fast, 'Mass/shift/bleed'
 - Brain parenchyma, ventricular system, bone
 - **4Hs: hyperdensity (blood), hypodensity (edema), hydrocephalus, herniation**
 - **95% sensitive for acute blood within 24 hours of symptom onset**
 - **Ischemia takes several hours to show up on CT**
- CT angiography: vascular pathology
 - Occlusion, stenosis, aneurysm, vasospasm
- MRI: better soft tissue discrimination
 - Small or early infarcts, **brainstem pathology**, diffuse axonal injury, microhemorrhages, characterize masses, gauge timing of ischemia or hemorrhage
 - Usually obtained in subacute period, **post stabilization**
 - Need clearance for **implanted devices or metallic foreign bodies prior to MRI**



○ Other testing

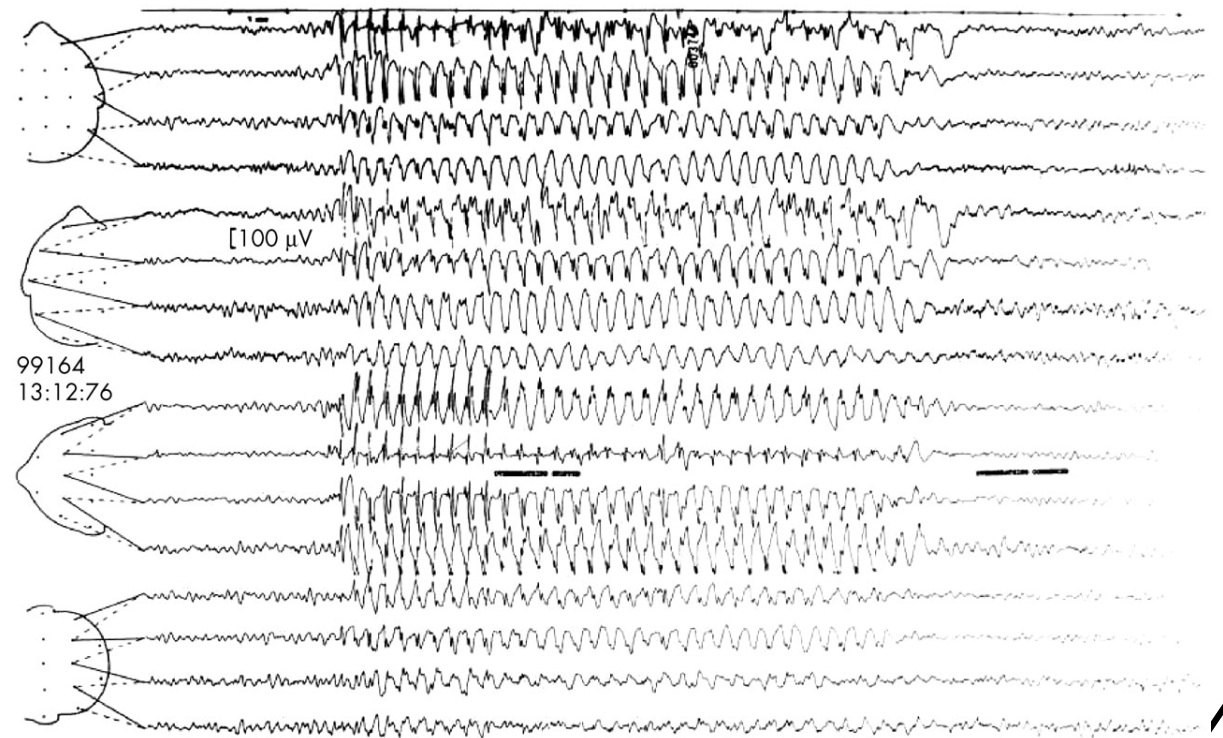
- Bloodwork and urine studies
- EEG*
- Lumbar puncture*
- EMG/NCS



○ EEG

- Indications:

- Monitoring in convulsive status epilepticus to ensure seizure control
- Monitoring burst suppression in patients with refractory status epilepticus
- Assessing for **nonconvulsive seizures or nonconvulsive status epilepticus** in patients with altered consciousness
- Clarifying etiology of abnormal movements and spells – differentiating between true seizures and pseudoseizures
- Adjunct tool in neurologic prognostication.



○ Lumbar Puncture

- Indications:

- Subacute, acute mental status change + fever/leukocytosis/signs of infection
- Unexplained seizures
- Unexplained confusion
- High suspicion for subarachnoid hemorrhage with negative CT scan (xanthochromia)
- Guillain Barre: cytoalbuminologic dissociation, antiganglioside antibodies
- Other autoimmune, inflammatory, or paraneoplastic CNS process

- Relative Contraindications

- Decreased LOC + focal neurologic deficit: **get a CT first** to assess elevated ICP and risk of herniation from LP.





Selected Neurological Emergencies

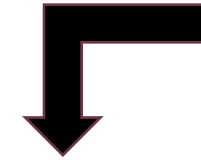




Acute ischemic stroke

- Covered by Dr. Massa
- Overview:
 - Account for **85% of all strokes**, **4th** leading cause of death in US
 - **"Time is Brain"** - prompt recognition/treatment essential to minimize permanent disability & mortality
- Pathophysiology
 - Occurs due to **diminished vascular supply to the brain**. This leads to an ischemic penumbra - tissue at risk of infarction that can be saved with timely reperfusion
 - Two types: **Embolic events** from cardiac sources (e.g., atrial fibrillation) or the carotid arteries. Or **Intracerebral atherosclerosis** affecting small vessels

acute



Chronic
(months)



- **Patient Presentation**
 - Characterized by the **acute onset of focal neurological deficits**.
 - **Determining the "last known well" time** is essential for treatment decisions.
 - Common signs include:
 - **Facial droop**
 - **Unilateral weakness or numbness**
 - **Aphasia** (language disturbance)
 - **Gaze deviation**
 - **Gait abnormalities**
 - The **National Institutes of Health Stroke Scale (NIHSS)** is used to quantify severity of stroke, and *serial exams* are crucial to monitor for changes.

<u>Item Number</u>	<u>Item Name</u>	<u>Scoring Guide</u>	<u>Patient Score</u>
1B	LOC Questions	0=answers both correctly 1=answers one correctly 2=answers neither correctly	_____
1C	LOC Commands	0=performs both tasks correctly 1=performs one task correctly 2=performs neither task	_____
2.	Gaze	0=normal 1=partial gaze palsey 2=total gaze palsey	_____
3.	Visual Fields	0=no visual loss 1=partial hemianopsia 2=complete hemianopsia 3=bilateral hemianopsia	_____
5a.	Left Arm Motor	0=no drift 1=drift before 10 seconds 2=falls before 10 seconds 3=no effort against gravity 4=no movement	_____
5b.	Right Arm Motor	0=no drift 1=drift before 10 seconds 2=falls before 10 seconds 3=no effort against gravity 4=no movement	_____
6a.	Left Leg Motor	0=no drift 1=drift before 5 seconds 2=falls before 5 seconds 3=no effort against gravity 4=no movement	_____
6b.	Right Leg Motor	0=no drift 1=drift before 5 seconds 2=falls before 5 seconds 3=no effort against gravity 4=no movement	_____
8.	Sensory	0=normal 1=abnormal	_____
9.	Language	0=normal 1=mild aphasia 2=severe aphasia 3=mute or global aphasia	_____
11.	Neglect	0=normal 1=mild 2=severe	_____

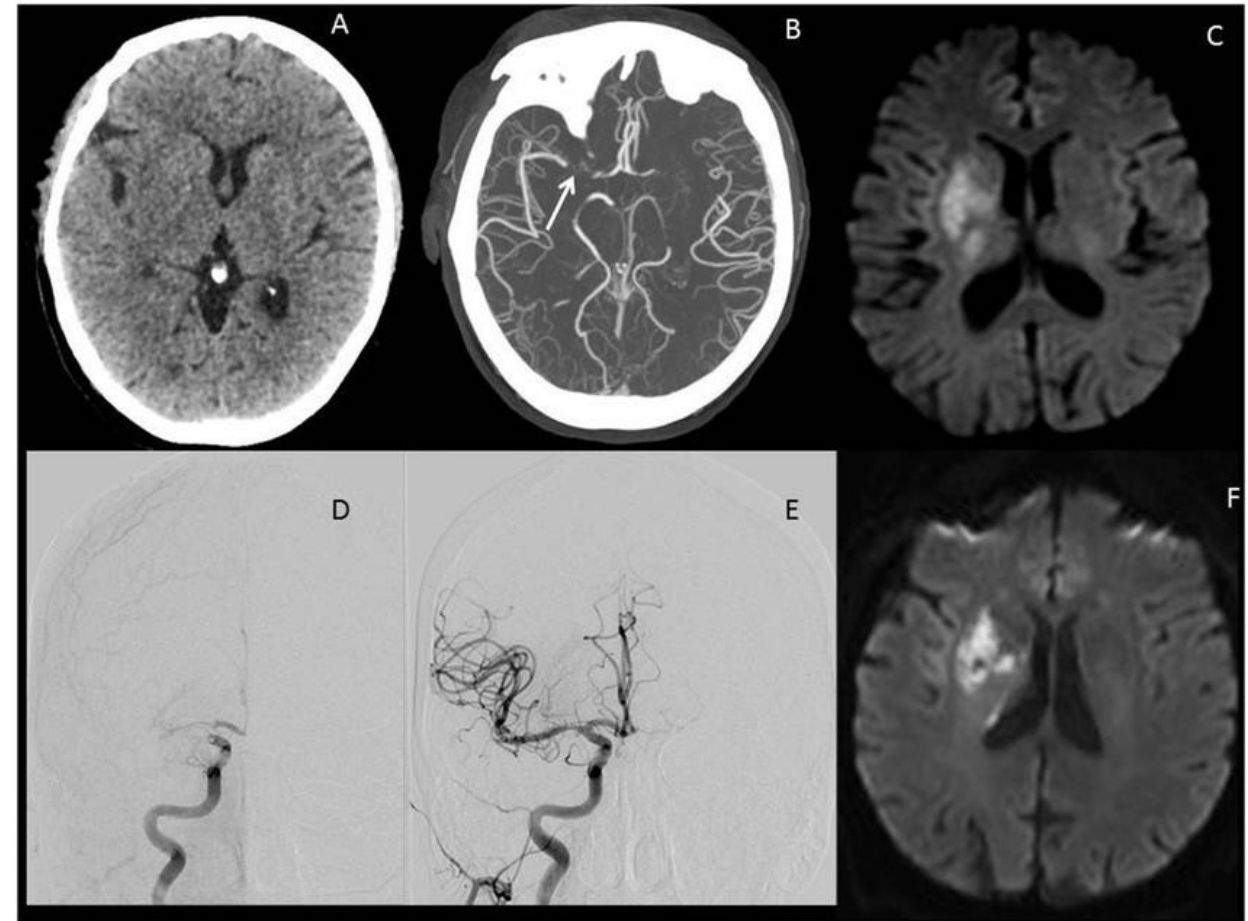
* Scoring from Original Scale

Score (out of 31): _____



• Gold Standard Diagnostic Testing

- **Emergent non-contrast CT of the head:** first imaging study performed to rule out hemorrhage
- **CT angiography (CTA)** immediately after to identify large vessel occlusion (LVO).
- **CT perfusion (CTP):** differentiates between infarct core (irreversibly damaged tissue) and ischemic penumbra (salvageable tissue) - guiding decisions for thrombectomy
- Initial lab work includes **blood glucose, coagulation studies (PT/INR, PTT), a complete blood count (CBC), and a basic metabolic panel**





• Basic Management

• Reperfusion Therapy:

- **Thrombolysis:** Intravenous tissue plasminogen activator (tPA) or tenecteplase (TNK) within a **3 to 4.5-hour window** from last known well (inclusion/exclusion criteria)
- **Endovascular Thrombectomy:** For LVO, mechanical clot retrieval up to **24 hours** from last known well in select patients

• Medical Management:

- Blood Pressure: **Permissive hypertension** (allowing SBP up to 220 mm Hg if not recanalized, or <185 mm Hg after reperfusion) to maintain perfusion to penumbra.
- **Blood Glucose:** 140-180 mg/dL, hypoglycemia & hyperglycemia associated with worse outcomes
- **Temperature:** Maintain normothermia - fever worsens neuronal injury

• Secondary Prevention:

- Aspirin + statin started 24 hours after thrombolysis
- Further workup includes echocardiography and telemetry to investigate the underlying cause (e.g., atrial fibrillation, carotid stenosis).



● Hemorrhagic Stroke

- Covered by Dr. Massa
- **Overview**
 - **15% of all strokes:** include bleeding into the brain parenchyma or surrounding spaces.
 - Main types relevant for emergencies:
 - **Intracerebral Hemorrhage (ICH):** Bleeding directly into brain tissue.
 - **Subarachnoid Hemorrhage (SAH):** Bleeding into cerebrospinal fluid (CSF)-containing spaces.
 - **Subdural Hematoma (SDH):** Bleeding into subdural space.
 - **Epidural Hematoma (EDH):** Bleeding into epidural space.

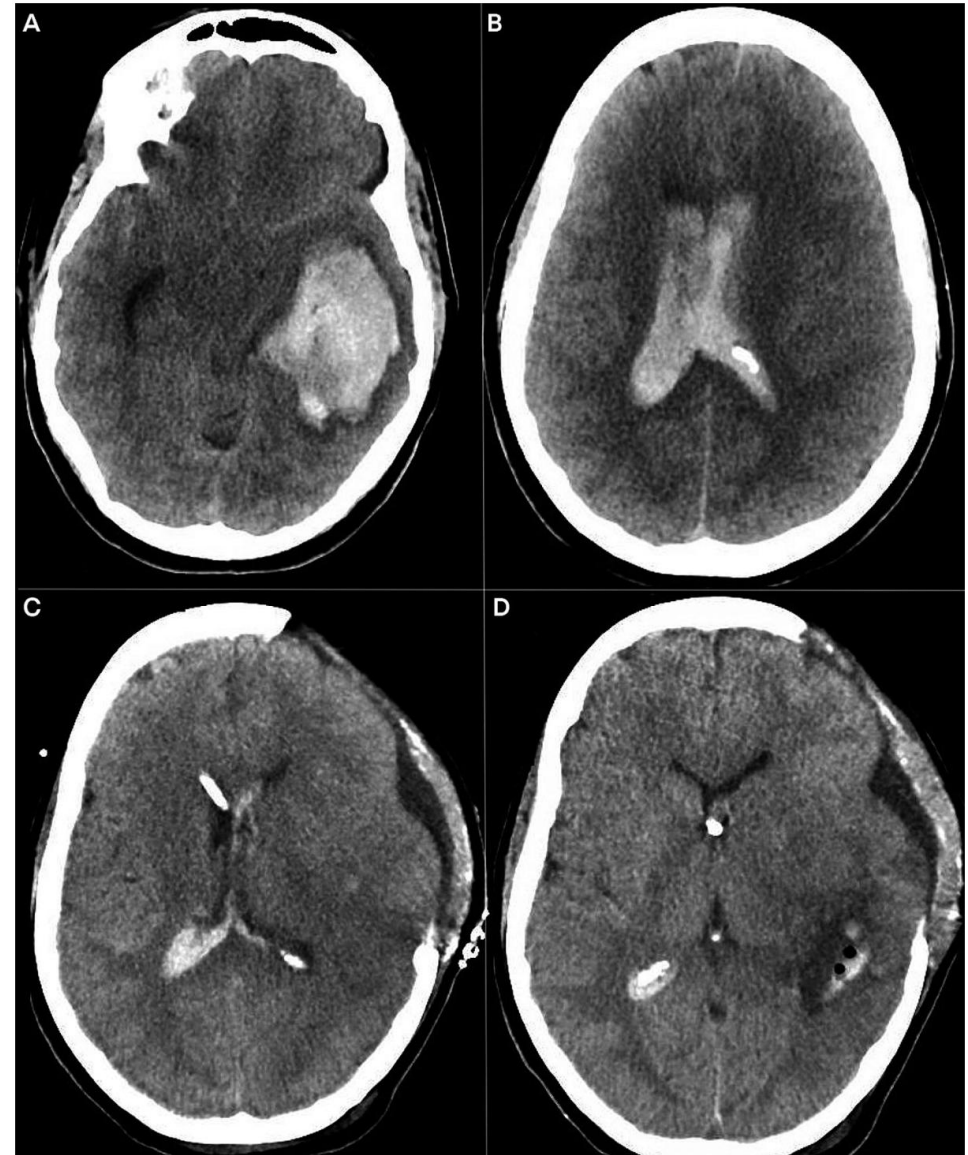


FIGURE 4-3

Imaging from the patient in **CASE 4-1**. A CT scan shows an acute intraparenchymal hemorrhage in the left temporal lobe (A) with concurrent intraventricular hemorrhage (B) and interval resolution of the burden of hematoma after surgical clot evacuation, decompressive hemicraniectomy, and placement of an external ventricular drain (C, D).



- Pathophysiology
 - **ICH**: due to **uncontrolled hypertension** (damaging deep perforating arteries) or **cerebral amyloid angiopathy** (affecting cortical vessels). Initial hematoma causes primary injury, secondary injury from inflammation and oxidative stress.
 - **SAH**: Most commonly caused by **rupture of a cerebral aneurysm**
 - **SDH**: Usually results from trauma causing tearing of **bridging veins** - *elderly and infants* are at higher risk.
 - **EDH**: Typically caused by head trauma leading to injury of the **middle meningeal artery**.
- Patient Presentation
 - **Similar symptoms as ischemic stroke. Imaging diagnosis.**
 - **ICH**: altered consciousness, headache, nausea/vomiting, severe hypertension, focal deficits, and potential for rapid deterioration. Hematoma expansion is a major concern, often occurring within the first 3-6 hours.
 - **SAH**: **thunderclap headache**, "worst headache of my life"

SDH



Figure 35-9. Acute subdural hematoma over the right convexity, with substantial mass effect (displacement) of brain tissue but little edema.

EDH

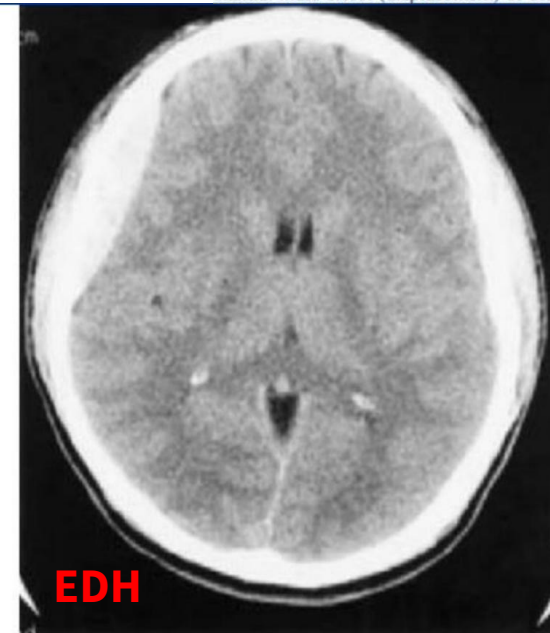


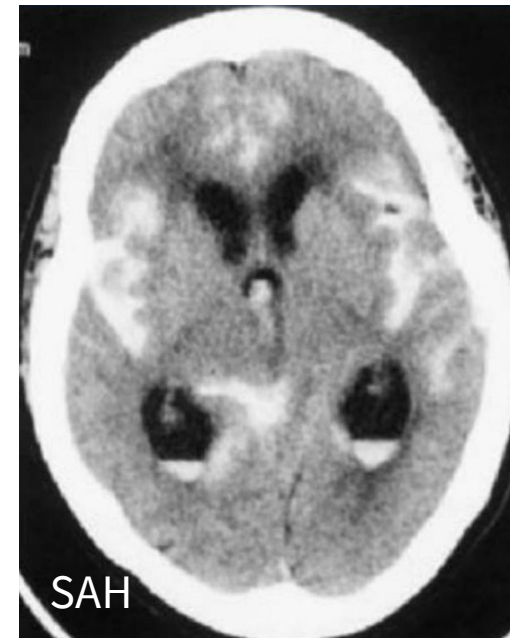
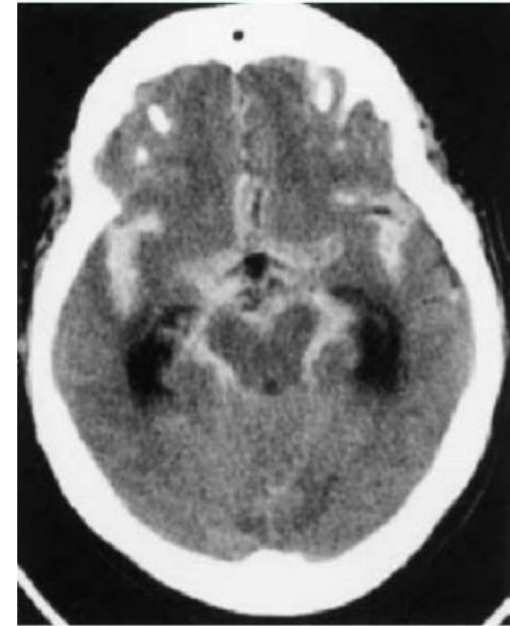
Figure 35-8. Acute epidural hematoma; unenhanced CT scan showing a typical lens-shaped frontal epidural clot.



• Gold Standard Diagnostic Testing



- **Non-contrast head CT** is the primary diagnostic tool.
 - For ICH, it is **98.6% sensitive** within 6 hours of onset.
 - For SAH, it is **95% sensitive** within the first 6 hours.
 - SDH appears as a **crescent-shaped** hemorrhage that crosses suture lines.
 - EDH appears as a **convex or lentiform-shaped** hemorrhage that does not cross suture lines.
- **Lumbar Puncture (LP)** is indicated for suspected SAH if the head CT is negative but clinical suspicion remains high. The presence of **xanthochromia** (yellow discoloration of CSF) is diagnostic.
- **CTA** is used to identify underlying vascular lesions like aneurysms in SAH or spot signs in ICH, which predict hematoma expansion.



- **ICH Management:**
 - **Aggressive blood pressure reduction** to a **SBP <140 mmHg within 3 hours** is recommended.
 - Often requires IV antihypertensives and continuous infusion
 - Avoid hypotension
 - **Reverse any coagulopathy** (e.g., due to warfarin or other anticoagulants) immediately.
 - Monitor for and treat elevated intracranial pressure (ICP).
 - ICP monitors not uncommon – EVDs, continuous EEG may be needed
 - Cerebellar hemorrhages may require **emergent neurosurgical evacuation**.

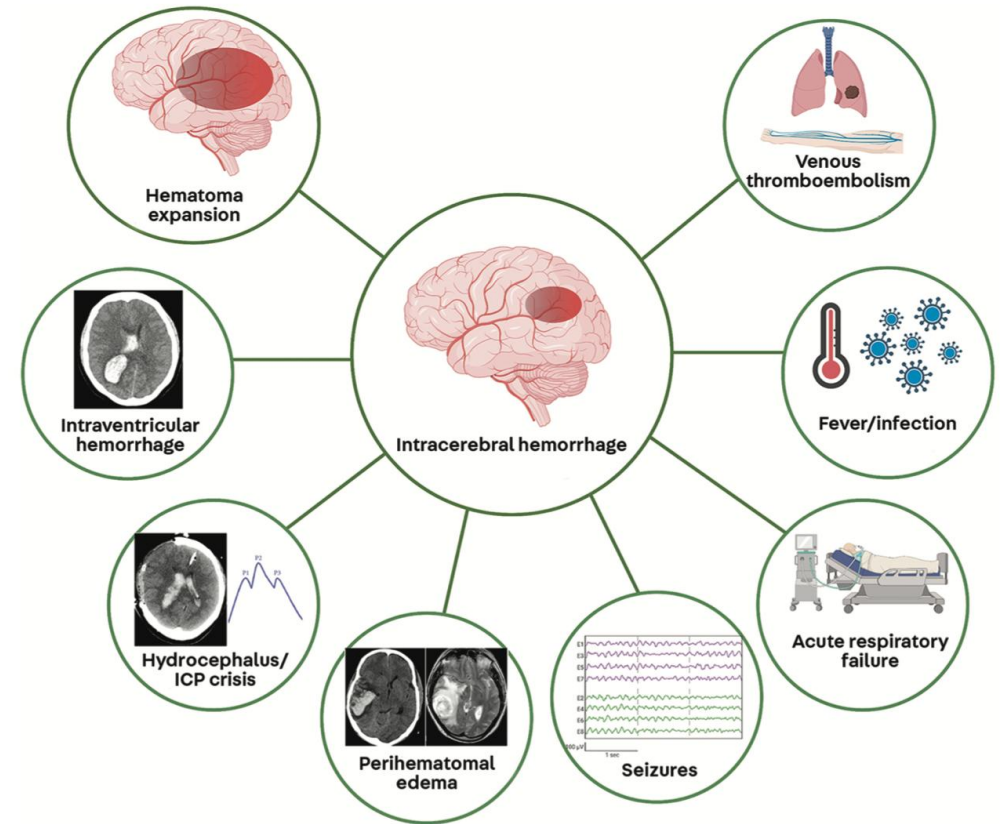


FIGURE 4-1

Summary of early complications after spontaneous intracerebral hemorrhage.

ICP = intracranial pressure.





• SAH Management:

- **Secure the aneurysm** via surgical clipping or endovascular coiling to prevent rebleeding – requires cerebral angiography
- Manage *blood pressure*, often more aggressively before the aneurysm is secured.
- Administer **nimodipine** to prevent delayed cerebral ischemia from vasospasm.
 - Presents with sudden change in neurological symptoms, can occur up to ~21 days after initial SAH
- Monitor for hydrocephalus and increased ICP, which may require an external ventricular drain (EVD).

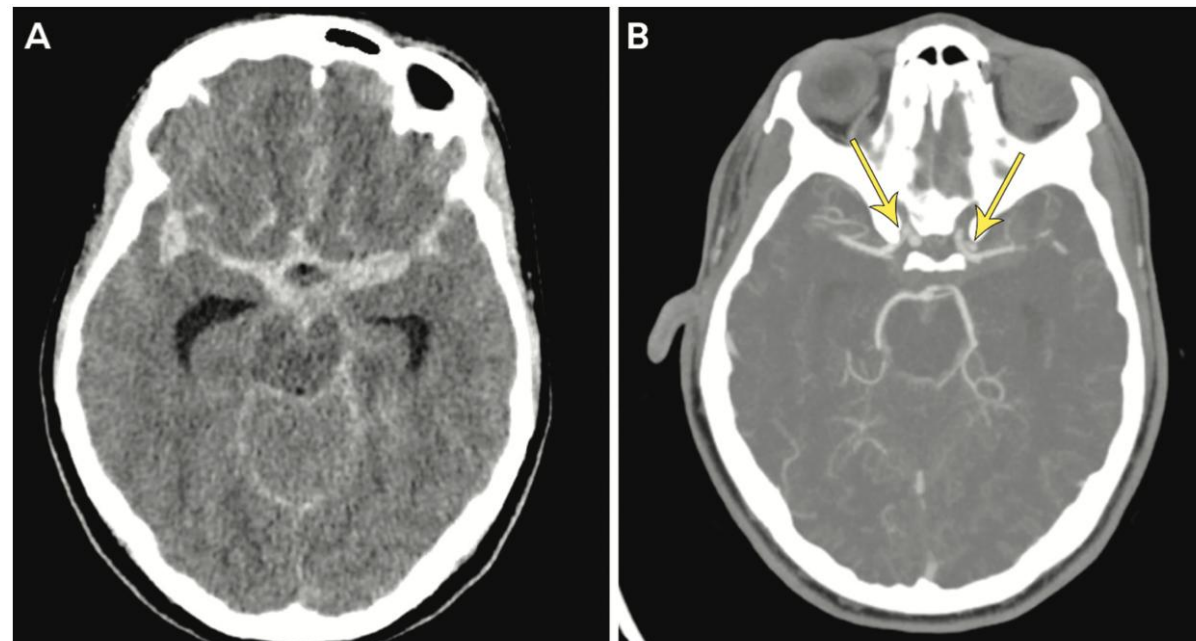


FIGURE 5-2

Imaging of the patient in **CASE 5-1**. A, Axial noncontrast head CT showing extensive subarachnoid blood and dilation of the temporal horns indicating hydrocephalus. B, Axial CT angiogram demonstrating multiple aneurysms in the circle of Willis (arrows).



○ Status Epilepticus

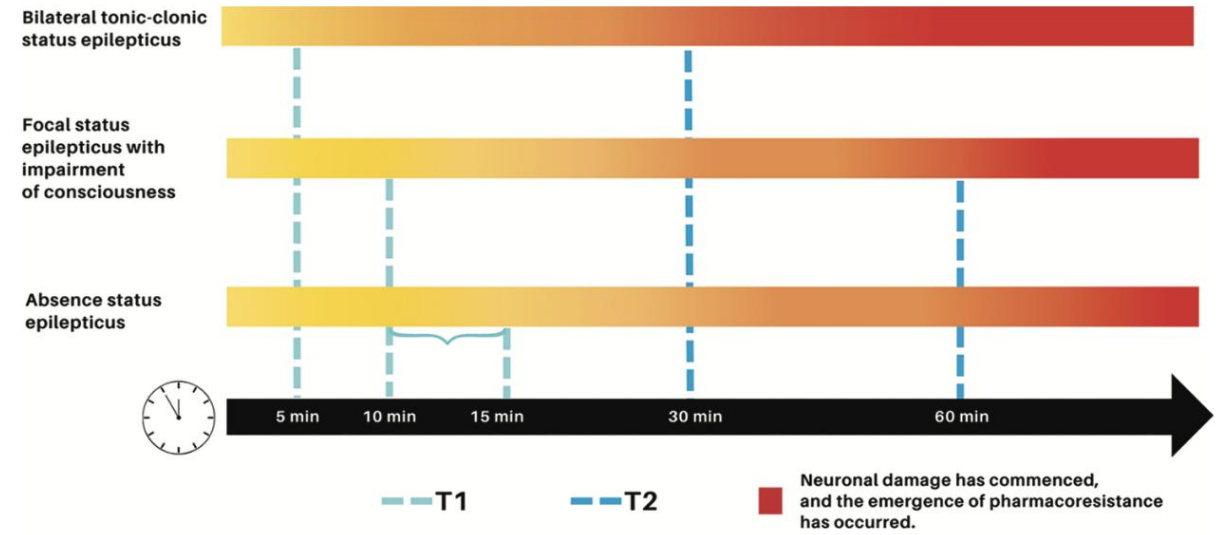
- **Definition and Pathophysiology**

- Continuous seizure activity or recurrent seizures without a return to baseline consciousness.

- The operational definition uses two time points:

- **T1**: Point at which a seizure unlikely to stop on its own (**5 minutes** for convulsive SE). **Treatment should start here.**
- **T2**: Long-term consequences, neuronal injury occurs (**30 minutes** for convulsive SE).

- Pathophysiology involves a failure of seizure termination mechanisms. **Downregulation of inhibitory GABA-A receptors** and increase in **excitatory NMDA receptors**, leads to pharmacoresistance, especially to benzodiazepines.



• Staging and Patient Presentation

- Increasing severity and treatment resistance:
 - **Stage 1 (Early)**: Seizures lasting >5 minutes.
 - **Stage 2 (Established)**: Seizures persist after initial benzodiazepine treatment.
 - **Stage 3 (Refractory)**: Seizures continue after a second-line antiseizure medication.
 - **Stage 4 (Super-refractory)**: Seizures persist for ≥ 24 hours after starting anesthetic agents.
- Presentation is classified by semiology:
 - **Convulsive SE**: Presents with prominent motor features, like generalized tonic-clonic movements.
 - **Non-convulsive SE (NCSE)**: Presents with altered mental status (confusion to coma), without motor convulsions. Subtle signs like facial twitching or nystagmus may be present. This is the most common type of SE in the ICU.

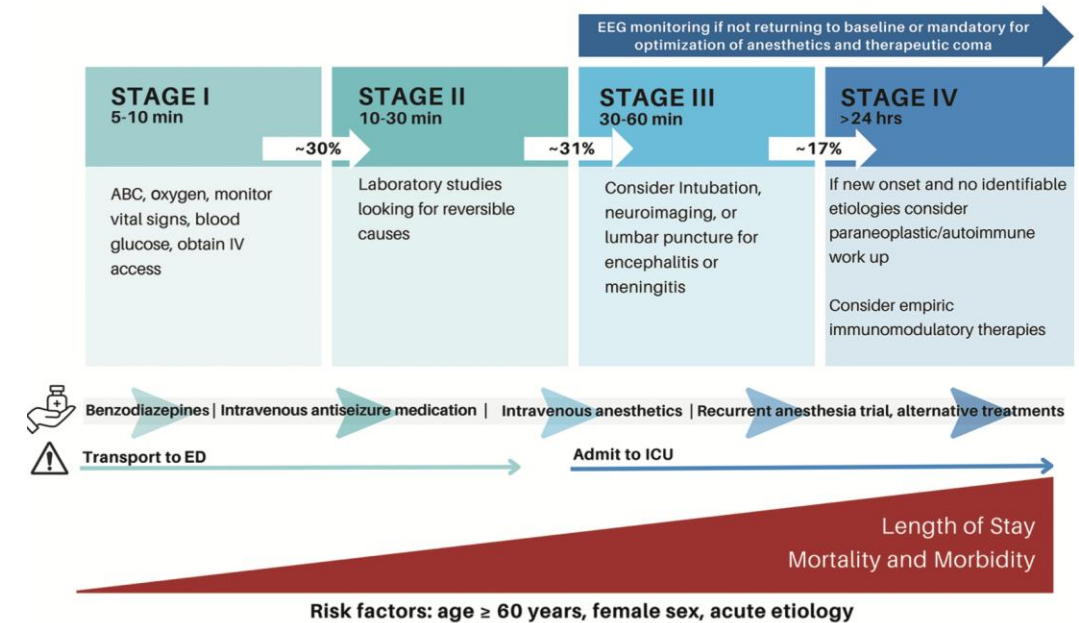


FIGURE 6-2

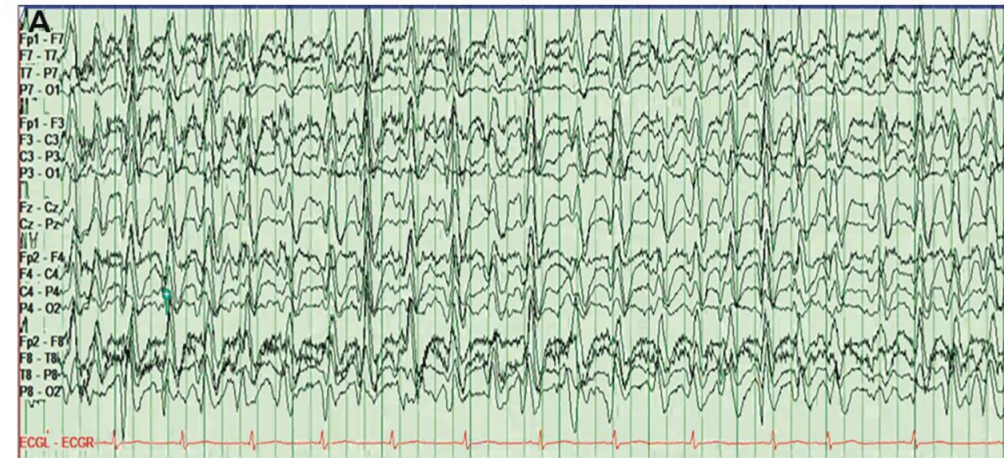
Clinical course and treatment stages of status epilepticus.

ABC = airway, breathing, circulation; ED = emergency department; ICU = intensive care unit.



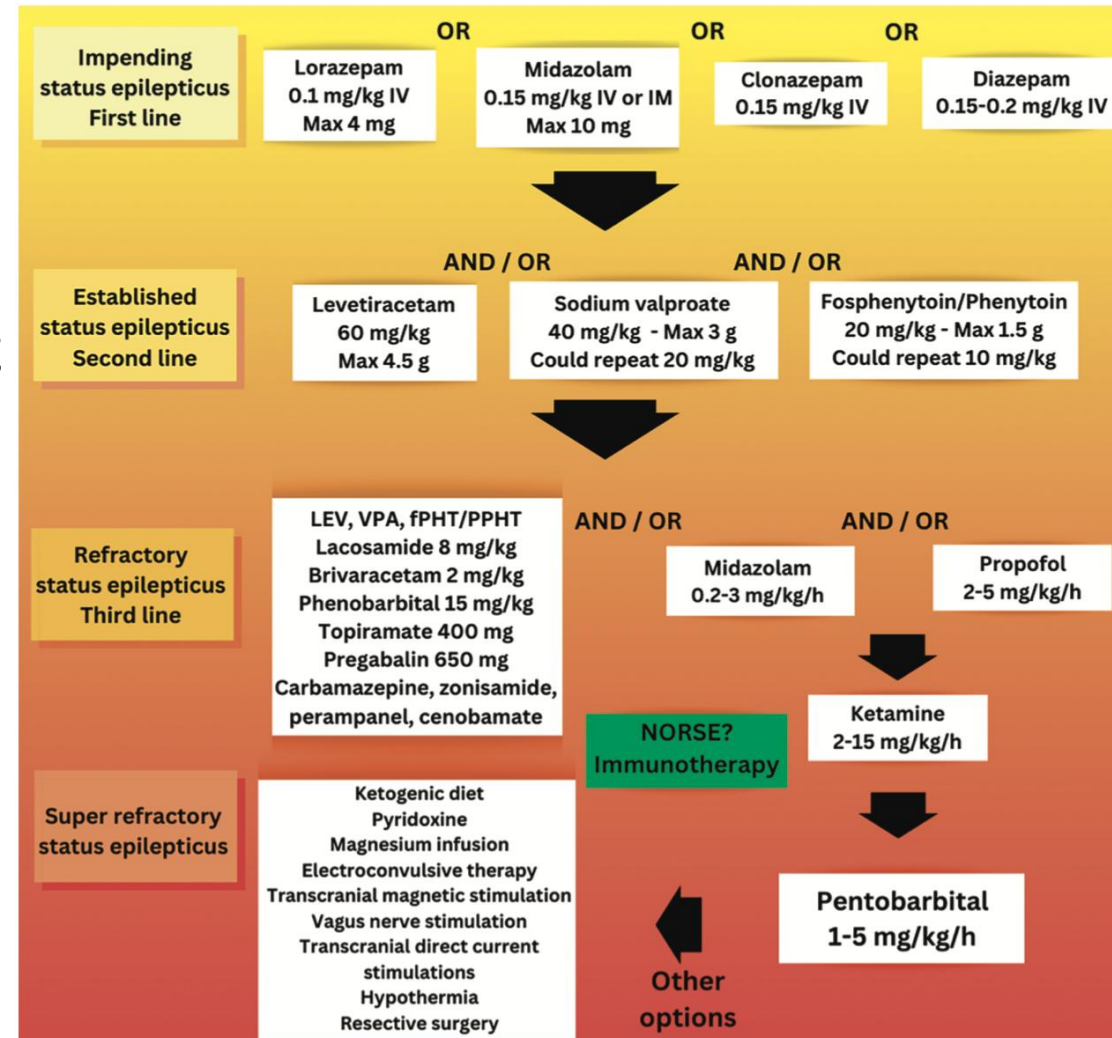
• Gold Standard Diagnostic Testing

- **Convulsive SE** is a **clinical diagnosis** based on **observation**
- **Electroencephalography (EEG)** is the gold standard for diagnosing **non-convulsive SE** and crucial in unexplained altered mental status after convulsion or in critically ill patients.
- Initial workup: blood glucose, electrolytes, toxicology screens, and levels of antiseizure medications to identify underlying cause.
- **Neuroimaging (CT or MRI)** is essential to look for acute structural causes like stroke, hemorrhage, or tumor.
- This EEG: A = SE, B = after benzos, C = resolution of periodic discharges



• Basic Management

- Goal: stop seizures as quickly as possible. **"Time is brain"**
- **Stage 1 (First-line):**
 - **Benzodiazepine** emergently: **IV lorazepam** is preferred, Intramuscular (IM) midazolam effective alternative if no IV access. Underdosing is a common cause of treatment failure.
- **Stage 2 (Second-line):**
 - Non-benzodiazepine IV antiseizure medication: fosphenytoin, valproic acid, or levetiracetam.
- **Stage 3 & 4 (Third-line/Refractory):**
 - Requires *admission to ICU* for continuous IV anesthetic infusions to induce **therapeutic coma**.
 - Common agents include **midazolam, propofol, ketamine, or pentobarbital**. Continuous EEG monitoring is required to guide therapy. Intubation is needed.



○ Spinal Cord Injury

- **Pathophysiology**

- **Primary Injury:** Initial mechanical trauma (fracture, compression, or laceration) - generally not reversible.
- **Secondary Injury:** Follows primary injury - ischemia, inflammation, and edema - worsens initial damage. Management focuses on mitigating this secondary injury.
- **Spinal Shock:** Transient state immediately after injury characterized by **flaccid paralysis and areflexia** below level of the lesion. This is distinct from neurogenic shock.

- **Patient Presentation and Initial Assessment**

- Suspect spinal cord injury (SCI) in any trauma patient. Initial management includes **spinal motion restriction** (e.g., cervical collar) to prevent further injury.
- Presentation depends on the level and completeness of the injury.
- **High cervical injuries (above C5)** often cause immediate **respiratory failure** due to paralysis of diaphragm and intercostal muscles, requiring intubation.
- **Injuries at or above T6** can cause **neurogenic shock**, a distributive shock state characterized by **hypotension and bradycardia** due to disruption of sympathetic outflow.



• Gold Standard Diagnostic Testing

- **CT scan** is the initial imaging modality of choice to evaluate for *bony fractures* and alignment issues.
- **MRI** is superior for visualizing the *spinal cord, ligaments, and intervertebral discs*. It is indicated in patients with neurologic deficits to assess for cord compression, contusion, hemorrhage, or ligamentous injury.
- **CTA** should be performed in patients with high-risk fractures (e.g., cervical spine) to screen for blunt cerebrovascular injury.

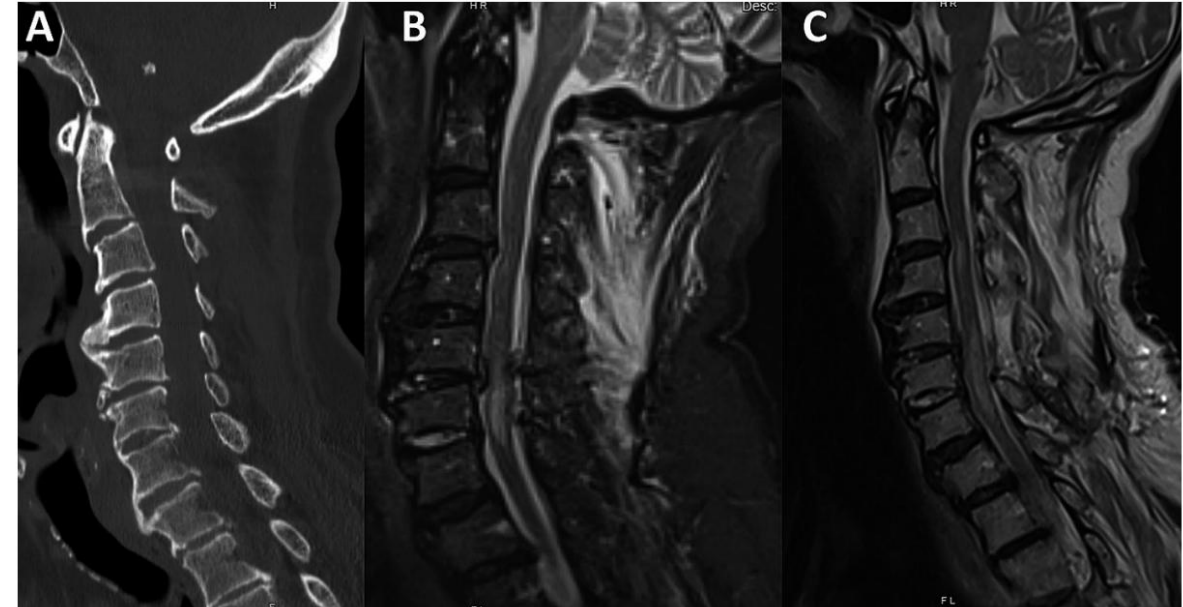


FIGURE 7-6

Imaging of the patient in **CASE 7-3**. Sagittal CT of the cervical spine (A) demonstrates degenerative changes with spinal canal stenosis and no acute fracture. Sagittal short tau inversion recovery (STIR) sequence MRI of the cervical spine (B) demonstrates cord contusion as a result of hyperextension injury at the C6 level and posterior soft tissue T2 hyperintensity indicative of ligamentous injury. Postoperative sagittal T2 MRI (C) shows expansion of the intramedullary lesion.





- Basic Management

- **Stabilization:** Airway, Breathing, and Circulation

- **Hemodynamic Support:**

- In neurogenic shock, initial treatment is fluid resuscitation, followed by vasopressors to maintain adequate perfusion.

- Current guidelines recommend maintaining a **mean arterial pressure of 85-90 mmHg for 7 days** to optimize spinal cord perfusion.

- **Surgical Decompression:**

- **Early surgical decompression** (within 24 hours) is associated with improved neurologic outcomes for patients with traumatic SCI.

- **Medical Management:**

- Corticosteroids are NOT recommended - lack of benefit, increased complications.

- Prophylaxis for venous thromboembolism, these patients are very high risk.

- Prevent pressure injuries with frequent repositioning and skin checks.



○ Neuromuscular emergencies

• Overview

- Disorders affecting **motor neurons, peripheral nerves, the neuromuscular junction, or muscles** that lead to **acute, rapidly progressive weakness**.
- Most life-threatening complication is **acute respiratory failure** due to weakness of the diaphragm and bulbar muscles. This typically manifests as hypercarbic (hypoventilation) respiratory failure. Hypoxia and hypercarbia are late signs.

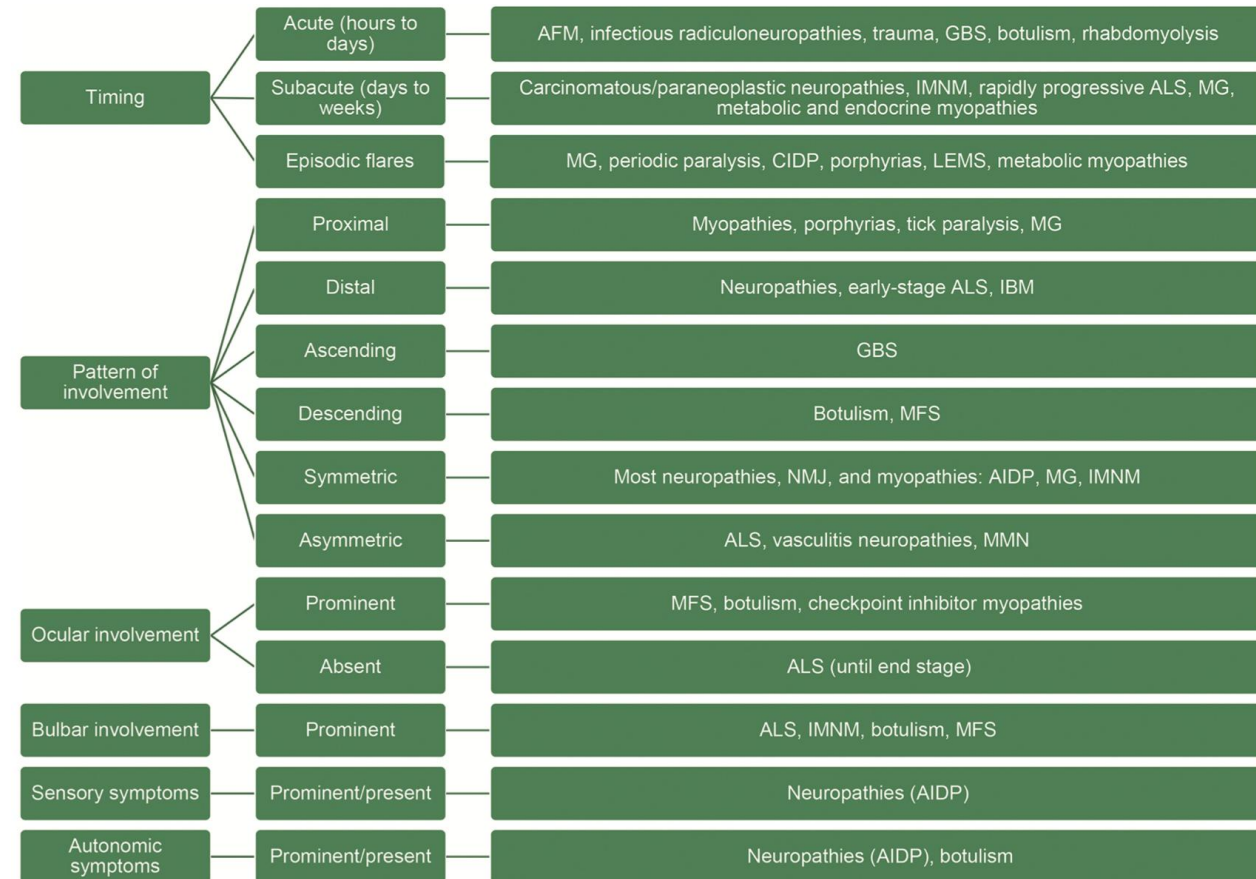


FIGURE 10-2

The evaluation of weakness and suggested differential diagnosis of neuromuscular emergencies.

AFM = acute flaccid myelitis; AIDP = acute inflammatory demyelinating polyradiculoneuropathy; ALS = amyotrophic lateral sclerosis; CIDP = chronic inflammatory demyelinating polyradiculoneuropathy; GBS = Guillain-Barré syndrome; IBM = inclusion body myositis; IMNM = immune-mediated necrotizing myopathy; LEMS = Lambert-Eaton myasthenic syndrome; MFS = Miller Fisher syndrome; MG = myasthenia gravis; MMN = multifocal motor neuropathy; NMJ = neuromuscular junction.



Exam findings in Neuromuscular emergencies

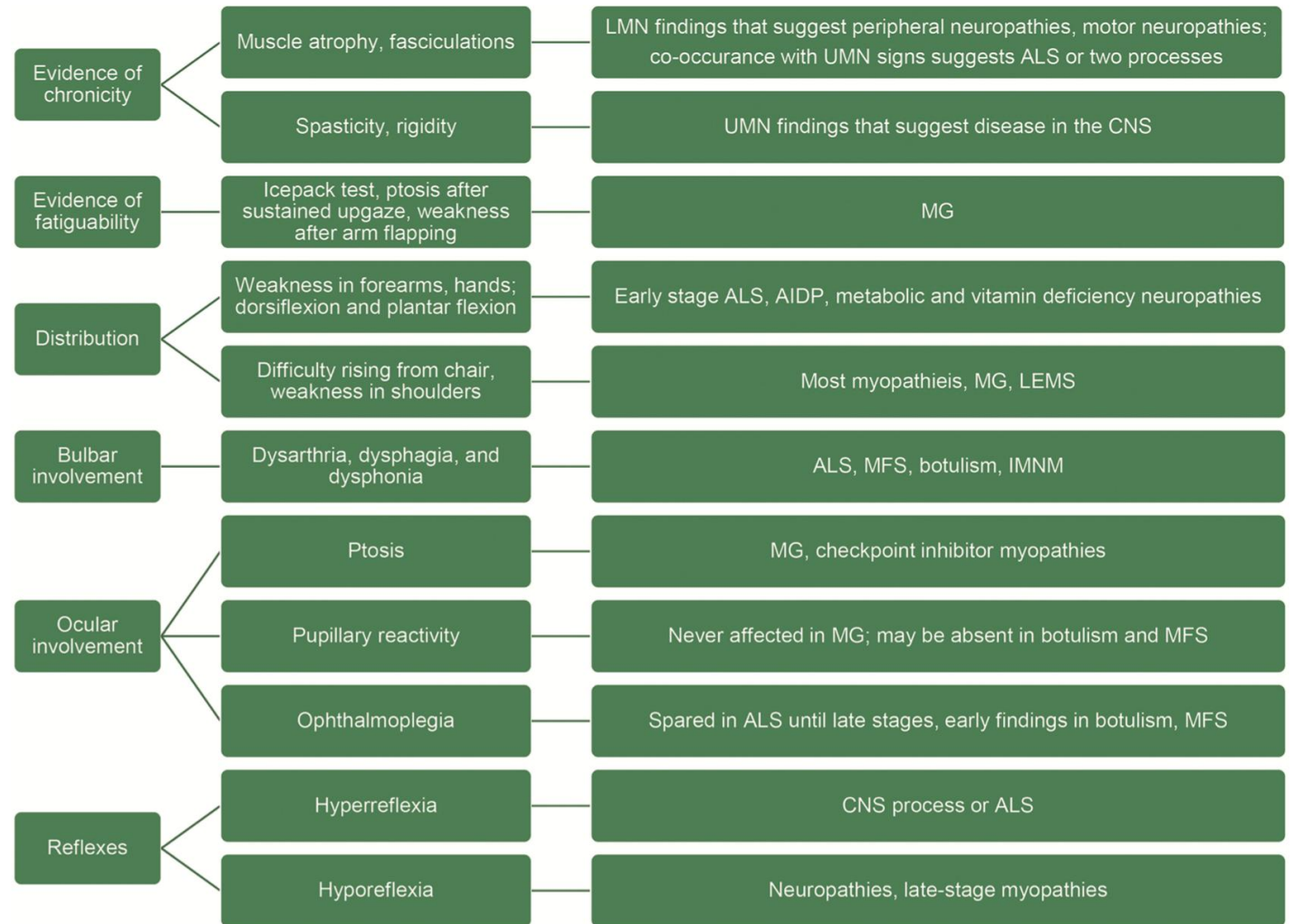


FIGURE 10-3

Examination maneuvers that refine the differential diagnosis in patients with suspected neuromuscular disease.

AIDP = acute inflammatory demyelinating polyradiculoneuropathy; ALS = amyotrophic lateral sclerosis; CNS = central nervous system; IMNM = immune-mediated necrotizing myopathy; LEMS = Lambert-Eaton myasthenic syndrome; LMN = lower motor neuron; MFS = Miller Fisher syndrome; MG = myasthenia gravis; UMN = upper motor neuron.



○ Guillain-Barré Syndrome (GBS)

- Pathophysiology and Presentation
 - **Pathophysiology:** GBS is a group of acute inflammatory polyradiculoneuropathies, often **preceded by an infection or vaccination**. The most common variant in the US is Acute Inflammatory Demyelinating Polyradiculoneuropathy (AIDP).
 - **Patient Presentation:**
 - Classic presentation is **rapidly progressive, symmetric ascending weakness**.
 - Sensory symptoms (paresthesias) are common.
 - **Areflexia or hyporeflexia** is a key finding, though reflexes may be normal early on.
 - **Autonomic dysfunction** (e.g., labile blood pressure, arrhythmias, urinary retention) occurs frequently.
 - **Bulbar weakness** (dysphagia, dysarthria) and facial weakness are concerning signs that predict respiratory failure.



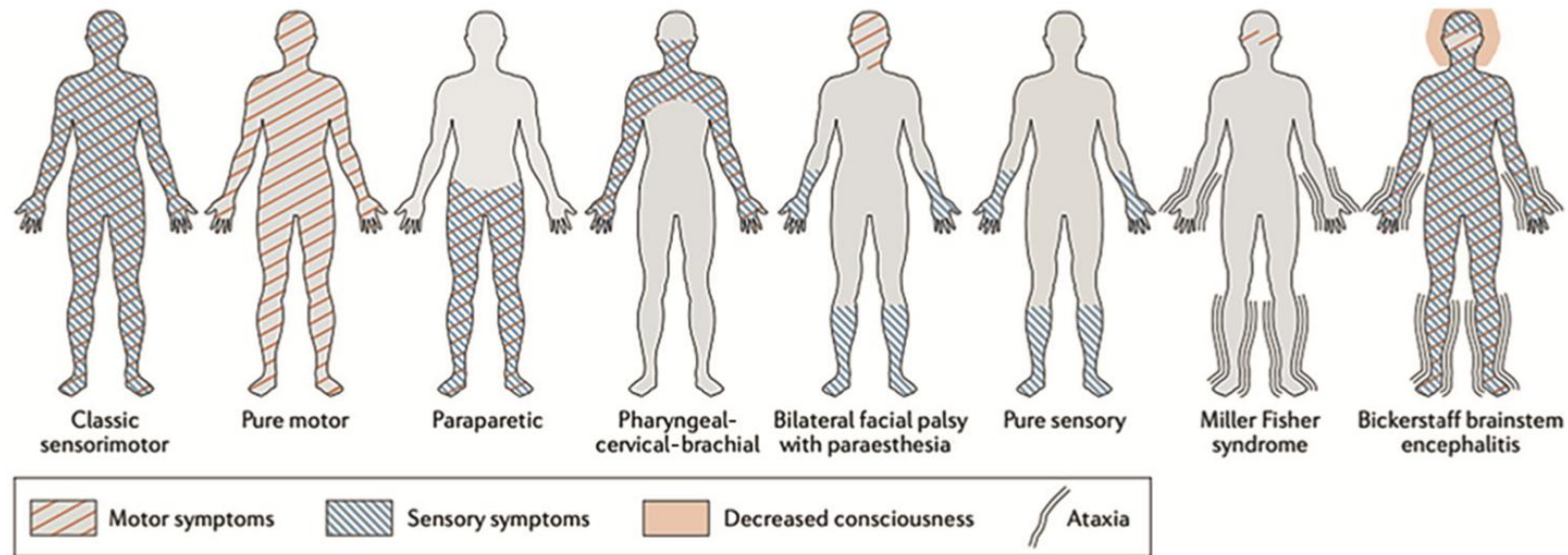


FIGURE 10-4

Variants of Guillain-Barré syndrome.

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Myasthenic Crisis

- Pathophysiology and Presentation
 - **Pathophysiology:** Myasthenia Gravis is an autoimmune disorder caused by antibodies against **postsynaptic acetylcholine receptors** (AChR). A **myasthenic crisis is an exacerbation** leading to respiratory failure.
 - **Patient Presentation:**
 - Hallmark is **fatigable weakness**, worsening with activity and improving with rest.
 - Often involves **oculobulbar muscles**, causing ptosis, diplopia, dysarthria, and dysphagia.
 - Crisis is often **triggered by infection, surgery, or certain medications**.
 - Important to distinguish from cholinergic crisis (weakness from too much pyridostigmine) - also presents with weakness but accompanied by excessive secretions and bradycardia.



● • **Gold Standard Diagnostic Testing**

- **Monitoring Respiratory Status:** Pulmonary Function Tests are critical
 - **Forced Vital Capacity (FVC)** and **Negative Inspiratory Force (NIF)** measured **serially** at bedside to track respiratory muscle strength
 - Intubation guided by **"20/30/40 rule"**: FVC <20 mL/kg, NIF less negative than -30 cm H₂O, or MEP <40 cm H₂O.
- **GBS Diagnosis:** Supported by **lumbar puncture showing cytoalbuminologic dissociation** (high protein, normal cell count) and nerve conduction studies showing demyelination.
- **MG Diagnosis:** Confirmed by **serum antibody testing (e.g., AChR antibodies)** and electrodiagnostic studies (repetitive nerve stimulation).





- Basic Management

- **Supportive Care:** Meticulous attention to respiratory status is paramount. Intubate proactively if there are signs of impending respiratory failure.
- **GBS Management:**
 - Treatment is with either **Intravenous Immunoglobulin (IVIg)** or **Plasma Exchange (PLEX)**. Both are equally effective.
 - **Steroids are NOT effective** for GBS.
- **MC Management:**
 - Treatment is also with **IVIg or PLEX**. PLEX may work faster in a crisis setting.
 - **High-dose steroids** are a cornerstone of treatment but may cause transient worsening, so they are often started after IVIg or PLEX is initiated.



○ Dr. Massa made me put this in here

- Movement Disorders Emergencies:
 - Oculogyric crisis – give Benadryl, don't wait for neurology
 - Sleep soundly



○ Brief intro to death by neurological criteria

- Determination that a patient meets **neurologic criteria for death** is a medical responsibility that leads to a **legal declaration of death**.
- Suspected in patients with *catastrophic brain injury*, no evidence of brain function, a state that must be ***permanent and irreversible***.
- Brain death examination **MUST** be accurate and consistent.
- Loss of brainstem's reticular activating system is most robust predictor of permanence of comatose state, neurologic assessment focuses **loss of all brainstem reflexes**
- **Criteria:** (1) coma, (2) brainstem areflexia, and (3) apnea in the setting of an adequate stimulus



○ My actual brain death note

Patient met pre-requisites to proceed with brain death testing:

- Coma, irreversible with known cause
- Neuroimaging explains coma
- CNS depressant drug effect absent
- No evidence of residual paralytics
- Absence of severe acid-base, electrolyte, endocrine abnormality
- Normothermia or mild hypothermia (core temp $>36^{\circ}\text{C}$)
- Systolic Blood Pressure greater than or equal to 100 mmHg
- No spontaneous respirations

Apnea Testing:

- Patient hemodynamically stable
- Ventilator adjusted to provide normocarbida, ABG pre-apnea test: pH *** PaCO₂ *** PaO₂ *** Bicarb ***
- Patient pre-oxygenated with 100% FiO₂ for 10 minutes to PaO₂ $>200\text{mmHg}$
- Patient well oxygenated with PEEP of 5cm of water
- Provided oxygen via suction catheter to the level of the carina at 6L/min or via T piece with CPAP at 10cm of water
- Disconnected ventilator
- Spontaneous respirations absent
- Arterial blood gas drawn at 8-10 minutes, patient reconnected to the ventilator
- PCO₂ $>60\text{mmHg}$ or 20mmHg rise from normal baseline value

Post apnea test ABG pH *** PaCO₂ *** PaO₂ *** Bicarb ***

Confirmatory Exam 2nd physician: ***

Time of Death ***

Examination:

- Pupils non reactive to bright light
- Corneal reflex absent
- Oculocephalic reflex absent
- Oculovestibular reflex absent
- No facial movement to noxious stimuli at supraorbital nerve, temporomandibular joint
- Gag reflex absent
- Cough reflex absent to tracheal suctioning
- Absence of motor response to noxious stimuli in all four limbs



○ Pearls

- There is a lot of overlap in exam findings for neurological emergencies, don't get overwhelmed, be systematic
- Your exam guides imaging and further work up
- Acute change in exam is a big deal, first sign of deterioration
- Final common pathway: injury → irritation → edema → elevated intracranial pressure → herniation → brain death





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